

Chapter 5

Link Layer

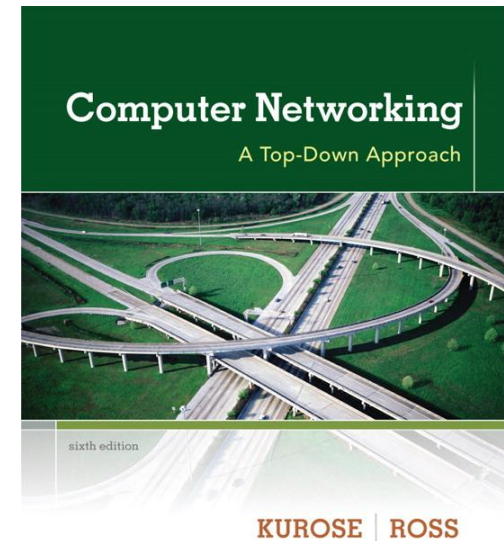
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**Computer
Networking: A Top
Down Approach**
6th edition
Jim Kurose, Keith Ross
Addison-Wesley
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Chapter 5: Link layer

our goals:

- ❖ understand principles behind link layer services:
 - error detection, correction
 - sharing a broadcast channel: multiple access
 - link layer addressing
 - local area networks: Ethernet, VLANs
- ❖ instantiation, implementation of various link layer technologies

Link layer, LANs: outline

5.1 introduction, services

5.2 error detection,
correction

5.3 multiple access
protocols

5.4 LANs

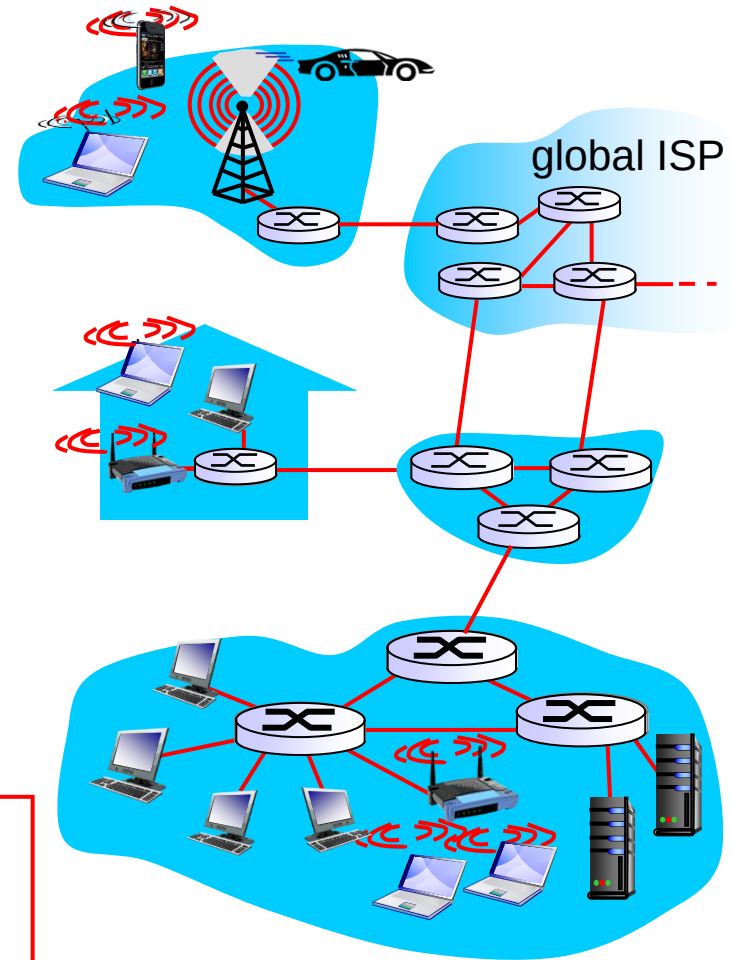
- addressing, ARP
- Ethernet

Link layer: introduction

terminology:

- ❖ hosts and routers: **nodes**
- ❖ communication channels that connect adjacent nodes along communication path: **links**
 - wired links
 - wireless links
 - LANs
- ❖ layer-2 packet: **frame**, encapsulates datagram

data-link layer has responsibility of transferring datagram from one node to *physically adjacent* node over a link



Link layer: context

- ❖ datagram transferred by different link protocols over different links:
 - e.g., Ethernet on first link, frame relay on intermediate links, 802.11 on last link
- ❖ each link protocol provides different services
 - e.g., may or may not provide rdt over link

transportation analogy:

- ❖ trip from Princeton to Lausanne
 - limo: Princeton to JFK
 - plane: JFK to Geneva
 - train: Geneva to Lausanne
- ❖ tourist = **datagram**
- ❖ transport segment = **communication link**
- ❖ transportation mode = **link layer protocol**
- ❖ travel agent = **routing algorithm**

Link layer services

❖ *framing, link access:*

- encapsulate datagram into frame, adding header, trailer
- channel access if shared medium
- “MAC” addresses used in frame headers to identify source, dest
 - different from IP address!

❖ *reliable delivery between adjacent nodes*

- we learned how to do this already (chapter 3)!
- seldom used on low bit-error link (fiber, some twisted pair)
- wireless links: high error rates
 - *Q*: why both link-level and end-end reliability?

Link layer services (more)

❖ *flow control:*

- pacing between adjacent sending and receiving nodes

❖ *error detection:*

- errors caused by signal attenuation, noise.
- receiver detects presence of errors:
 - signals sender for retransmission or drops frame

❖ *error correction:*

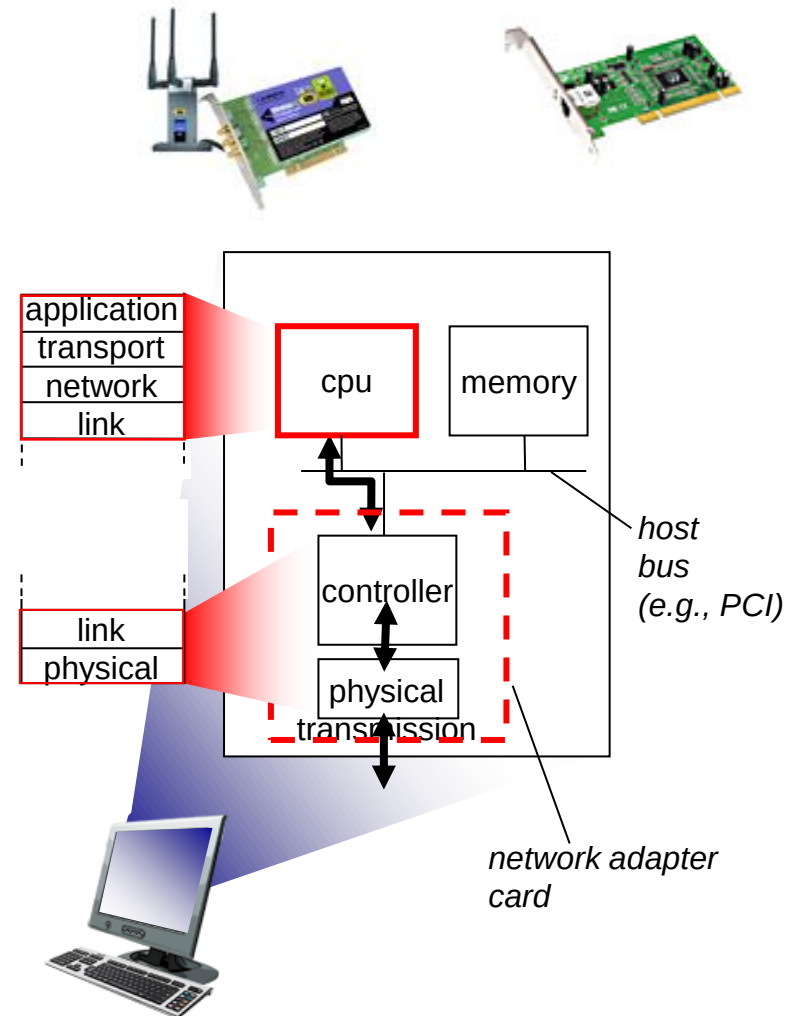
- receiver identifies *and corrects* bit error(s) without resorting to retransmission

❖ *half-duplex and full-duplex*

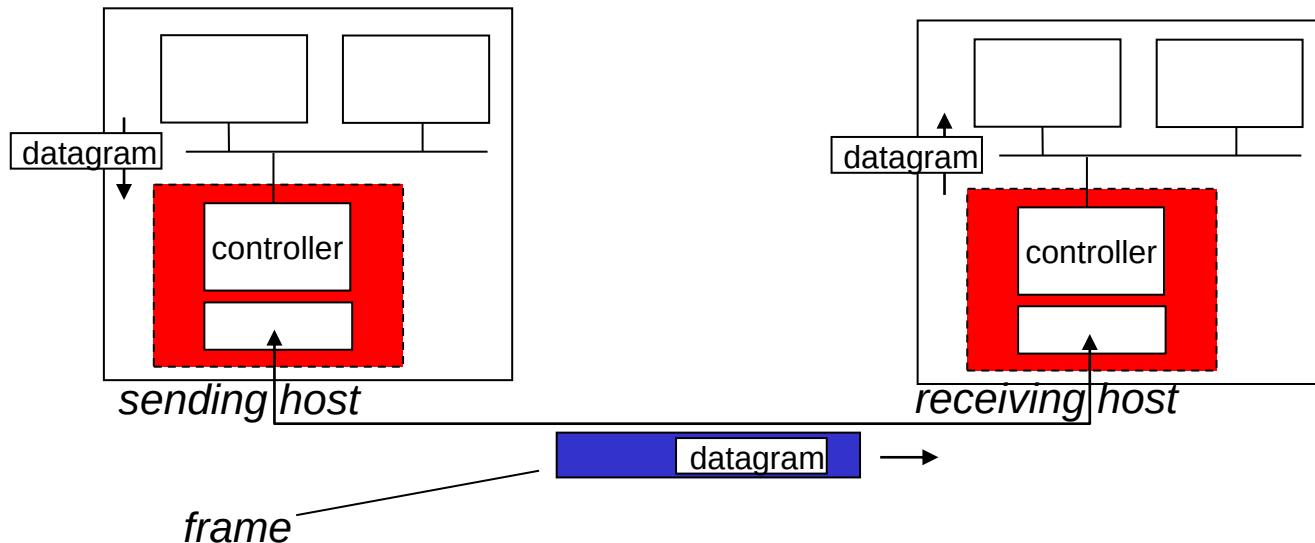
- with half duplex, nodes at both ends of link can transmit, but not at same time

Where is the link layer implemented?

- ❖ in each and every host
- ❖ link layer implemented in “adaptor” (aka *network interface card* NIC) or on a chip
 - Ethernet card, 802.11 card; Ethernet chipset
 - implements link, physical layer
- ❖ attaches into host's system buses
- ❖ combination of hardware, software, firmware



Adaptors communicating



❖ sending side:

- encapsulates datagram in frame
- adds error checking bits, rdt, flow control, etc.

❖ receiving side

- looks for errors, rdt, flow control, etc
- extracts datagram, passes to upper layer at receiving side

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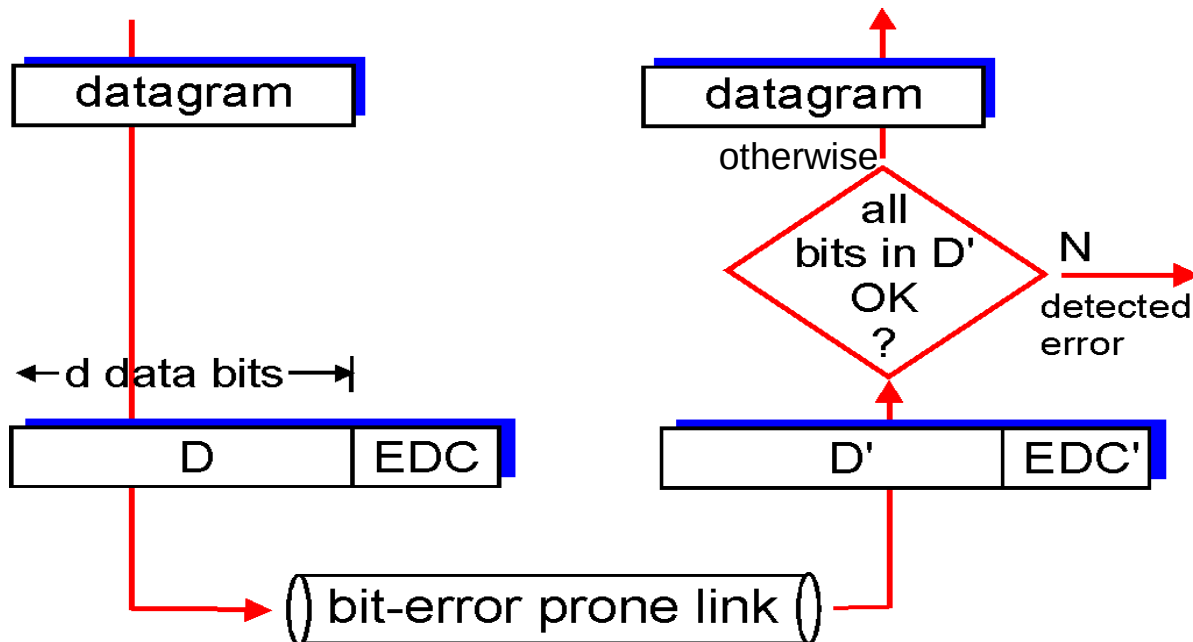
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Error detection

EDC= Error Detection and Correction bits (redundancy)

D = Data protected by error checking, may include header fields

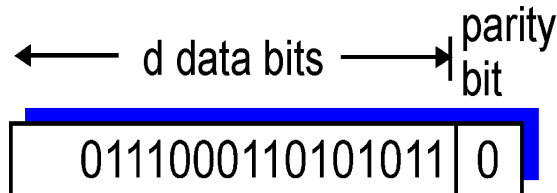
- Error detection not 100% reliable!
 - protocol may miss some errors, but rarely
 - larger EDC field yields better detection and correction



Parity checking

single bit parity:

- ❖ detect single bit errors



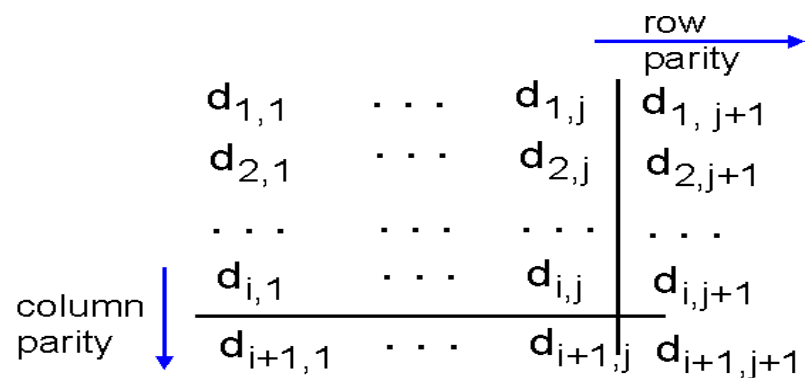
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Even Parity Scheme

Odd Parity Scheme

two-dimensional bit parity:

- ❖ detect and correct single bit errors



1	0	1	0	1	1
1	1	1	1	0	0
0	1	1	1	0	1
0	0	1	0	1	0

no errors

1	0	1	0	1	1
1	1	1	1	0	0
0	1	1	1	0	1
0	0	1	0	1	0

parity error

*correctable
single bit error*

Internet checksum (review)

goal: detect “errors” (e.g., flipped bits) in transmitted packet
(note: used at transport layer *only*)

sender:

- ❖ treat segment contents as sequence of 16-bit integers
- ❖ checksum: addition (1's complement sum) of segment contents
- ❖ sender puts checksum value into UDP checksum field

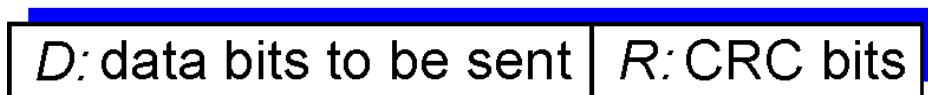
receiver:

- ❖ compute checksum of received segment
- ❖ check if computed checksum equals checksum field value:
 - NO - error detected
 - YES - no error detected.
But maybe errors nonetheless?

Cyclic redundancy check

- ❖ more powerful error-detection coding
- ❖ view data bits, **D**, as a binary number
- ❖ choose $r+1$ bit pattern (generator), **G**
- ❖ goal: choose r CRC bits, **R**, such that
 - $\langle D, R \rangle$ exactly divisible by G (modulo 2)
 - receiver knows G , divides $\langle D, R \rangle$ by G . If non-zero remainder: error detected!
 - can detect all burst errors less than $r+1$ bits
- ❖ widely used in practice (Ethernet, 802.11 WiFi, ATM)

← d bits → ← r bits →



*bit
pattern*

$$D * 2^r \text{ XOR } R$$

*mathematical
formula*

CRC example

want:

$$D \cdot 2^r \text{ XOR } R = nG$$

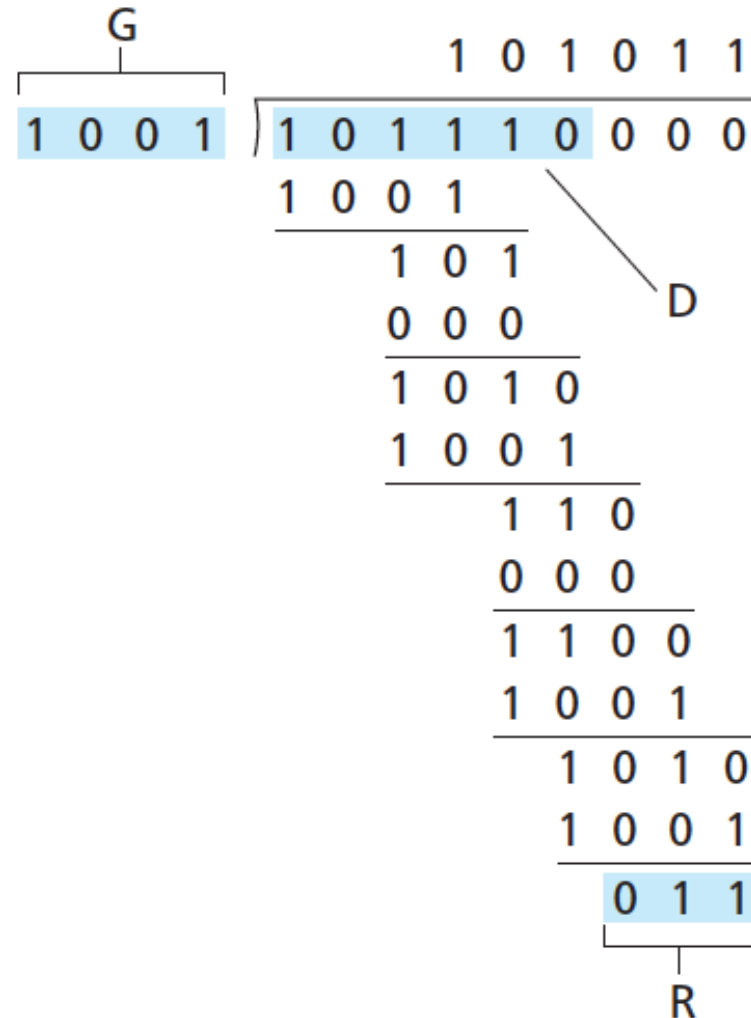
equivalently:

$$D \cdot 2^r = nG \text{ XOR } R$$

equivalently:

if we divide $D \cdot 2^r$ by G , want remainder R to satisfy:

$$R = \text{remainder}\left[\frac{D \cdot 2^r}{G}\right]$$



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- Ethernet

Multiple access links, protocols

two types of “links”:

- ❖ point-to-point

- PPP for dial-up access
- point-to-point link between Ethernet switch, host

- ❖ *broadcast (shared wire or medium)*

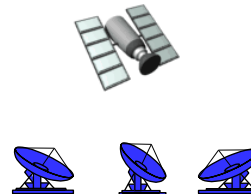
- old-fashioned Ethernet
- upstream HFC
- 802.11 wireless LAN



shared wire (e.g.,
cabled Ethernet)



shared RF
(e.g., 802.11 WiFi)



shared RF
(satellite)



humans at a
cocktail party
(shared air, acoustical)

Multiple access protocols

- ❖ single shared broadcast channel
- ❖ two or more simultaneous transmissions by nodes: interference
 - *collision* if node receives two or more signals at the same time

multiple access protocol

- ❖ distributed algorithm that determines how nodes share channel, i.e., determine when node can transmit
- ❖ communication about channel sharing must use channel itself!
 - no out-of-band channel for coordination

An ideal multiple access protocol

given: broadcast channel of rate R bps

desiderata:

1. when one node wants to transmit, it can send at rate R .
2. when M nodes want to transmit, each can send at average rate R/M
3. fully decentralized:
 - no special node to coordinate transmissions
 - no synchronization of clocks, slots
4. simple

MAC protocols: taxonomy

three broad classes:

- ❖ *channel partitioning*

- divide channel into smaller “pieces” (time slots, frequency, code)
- allocate piece to node for exclusive use

- ❖ *random access*

- channel not divided, allow collisions
- “recover” from collisions

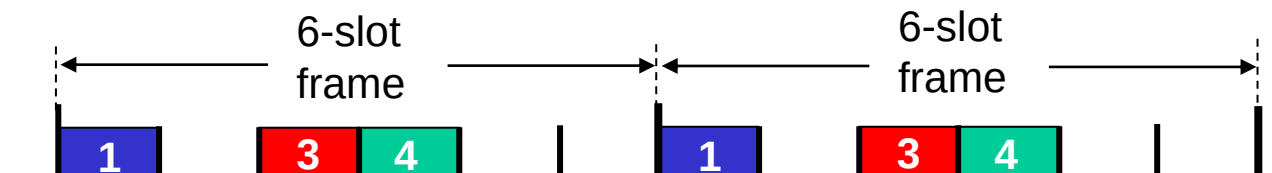
- ❖ *“taking turns”*

- nodes take turns, but nodes with more to send can take longer turns

Channel partitioning MAC protocols: TDMA

TDMA: time division multiple access

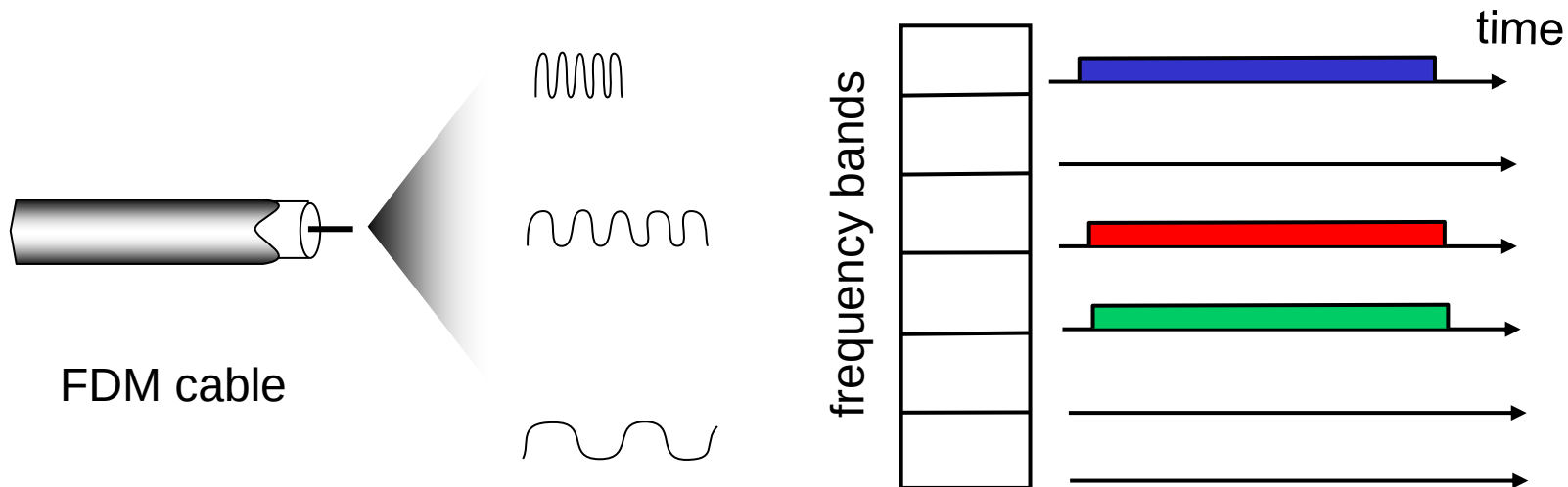
- ❖ access to channel in "rounds"
- ❖ each station gets fixed length slot (length = pkt trans time) in each round
- ❖ unused slots go idle
- ❖ example: 6-station LAN, 1,3,4 have pkt, slots 2,5,6 idle



Channel partitioning MAC protocols: FDMA

FDMA: frequency division multiple access

- ❖ channel spectrum divided into frequency bands
- ❖ each station assigned fixed frequency band
- ❖ unused transmission time in frequency bands go idle
- ❖ example: 6-station LAN, 1,3,4 have pkt, frequency bands 2,5,6 idle



Random access protocols

- ❖ when node has packet to send
 - transmit at full channel data rate R .
 - no *a priori* coordination among nodes
- ❖ two or more transmitting nodes → “collision”,
- ❖ **random access MAC protocol** specifies:
 - how to detect collisions
 - how to recover from collisions (e.g., via delayed retransmissions)
- ❖ examples of random access MAC protocols:
 - slotted ALOHA
 - ALOHA
 - CSMA, CSMA/CD, CSMA/CA

Slotted ALOHA

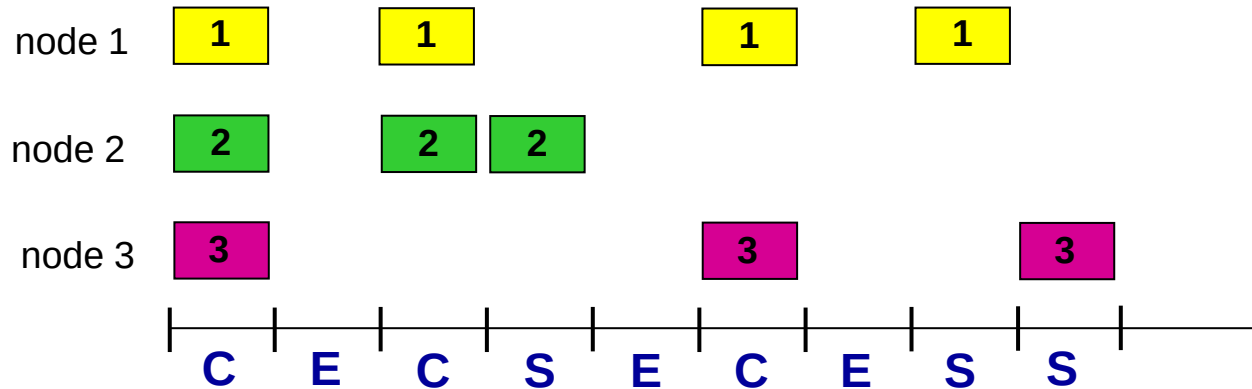
assumptions:

- ❖ all frames same size
- ❖ time divided into equal size slots (time to transmit 1 frame)
- ❖ nodes start to transmit only slot beginning
- ❖ nodes are synchronized
- ❖ if 2 or more nodes transmit in slot, all nodes detect collision

operation:

- ❖ when node obtains fresh frame, transmits in next slot
 - *if no collision:* node can send new frame in next slot
 - *if collision:* node retransmits frame in each subsequent slot with prob. p until success

Slotted ALOHA



Pros:

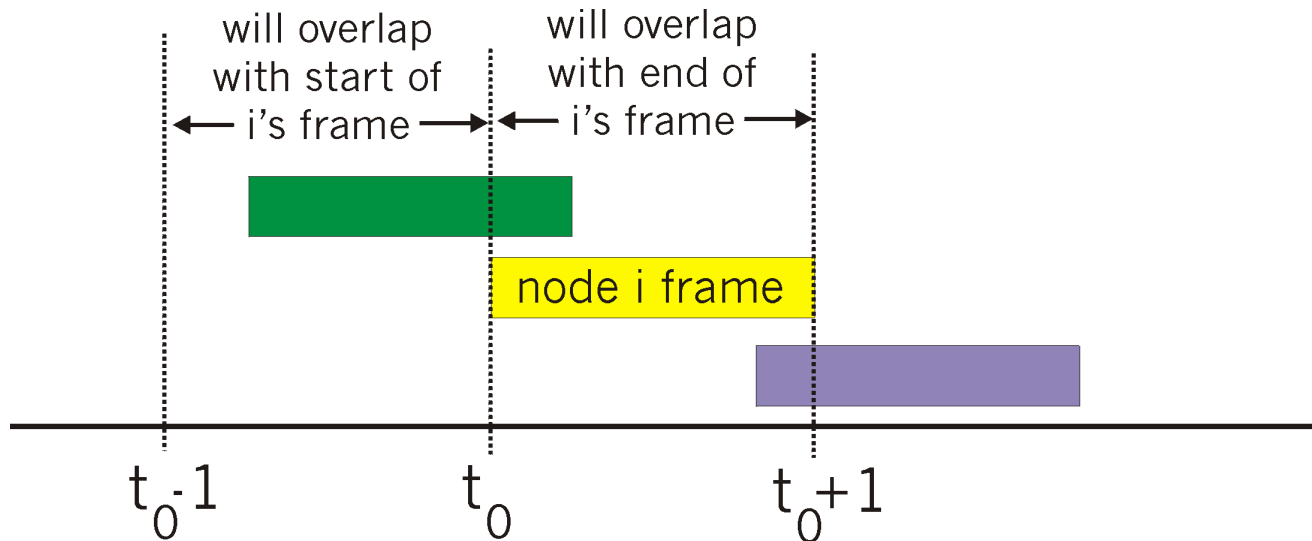
- ❖ single active node can continuously transmit at full rate of channel
- ❖ highly decentralized: only slots in nodes need to be in sync
- ❖ simple

Cons:

- ❖ collisions, wasting slots
- ❖ idle slots
- ❖ nodes may be able to detect collision in less than time to transmit packet
- ❖ clock synchronization

Pure (unslotted) ALOHA

- ❖ unslotted Aloha: simpler, no synchronization
- ❖ when frame first arrives
 - transmit immediately
- ❖ collision probability increases:
 - frame sent at t_0 collides with other frames sent in $[t_0 - 1, t_0 + 1]$



CSMA (carrier sense multiple access)

CSMA: listen before transmit:

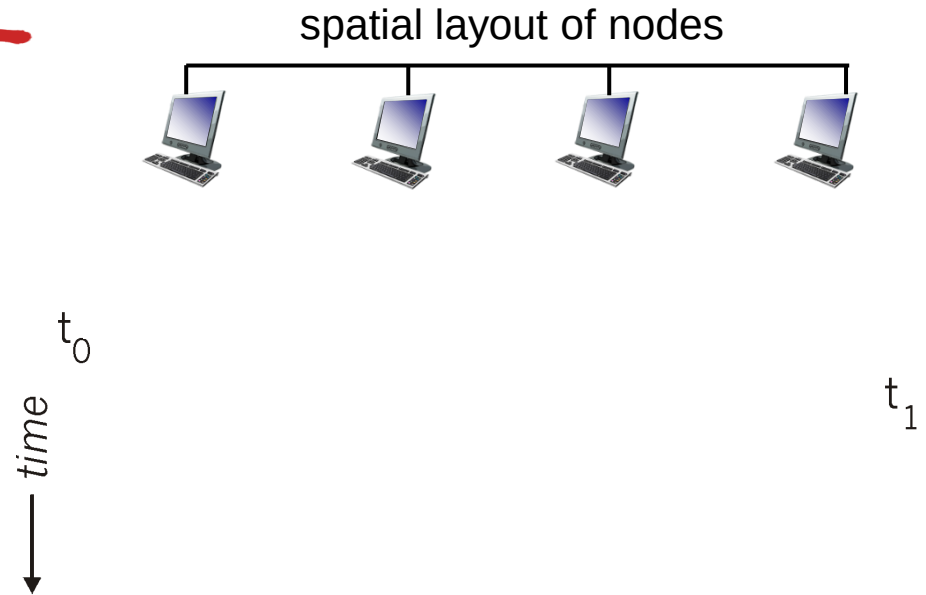
if channel sensed idle: transmit entire frame

❖ if channel sensed busy, defer transmission

❖ human analogy: don't interrupt others!

CSMA collisions

- ❖ collisions *can* still occur:
propagation delay means
two nodes may not hear
each other's
transmission
- ❖ collision: entire packet
transmission time
wasted
 - distance & propagation
delay play role in in
determining collision
probability

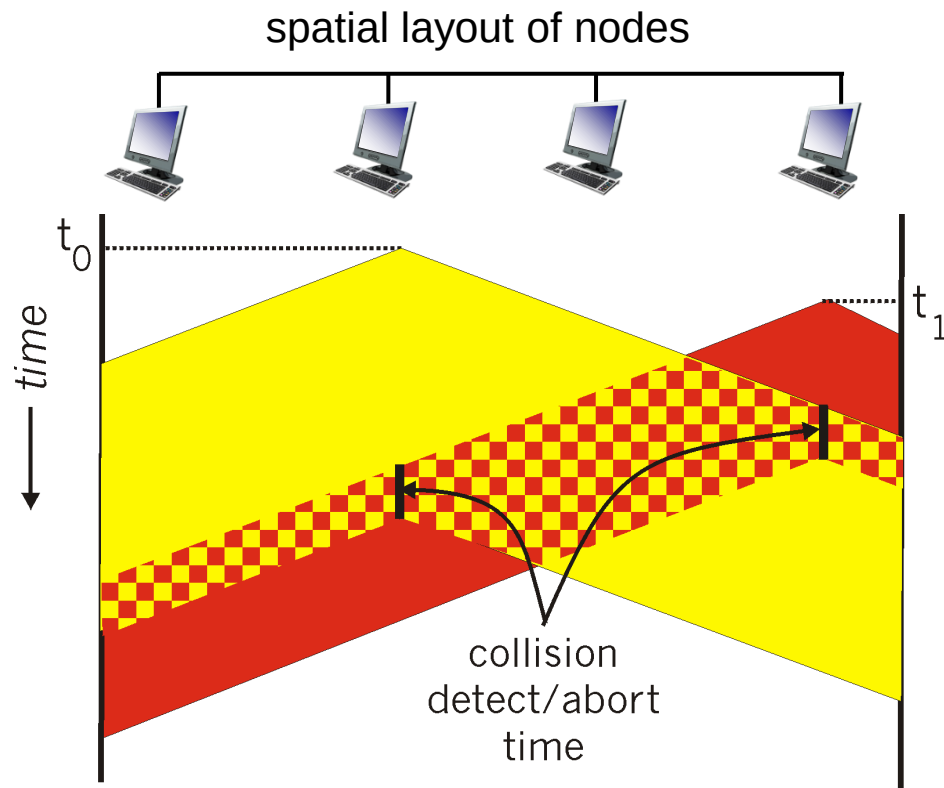


CSMA/CD (collision detection)

CSMA/CD: carrier sensing, deferral as in CSMA

- collisions *detected* within short time
- colliding transmissions aborted, reducing channel wastage
- ❖ collision detection:
 - easy in wired LANs: measure signal strengths, compare transmitted, received signals
 - difficult in wireless LANs: received signal strength overwhelmed by local transmission strength
- ❖ human analogy: the polite conversationalist

CSMA/CD (collision detection)



CSMA/CD efficiency

- ❖ T_{prop} = max prop delay between 2 nodes in LAN
- ❖ t_{trans} = time to transmit max-size frame

$$\text{efficiency} = \frac{1}{1 + 5t_{\text{prop}}/t_{\text{trans}}}$$

- ❖ efficiency goes to 1
 - as t_{prop} goes to 0
 - as t_{trans} goes to infinity
- ❖ better performance than ALOHA: and simple, cheap, decentralized!

“Taking turns” MAC protocols

channel partitioning MAC protocols:

- share channel *efficiently* and *fairly* at high load
- inefficient at low load: delay in channel access, 1/N bandwidth allocated even if only 1 active node!

random access MAC protocols

- efficient at low load: single node can fully utilize channel
- high load: collision overhead

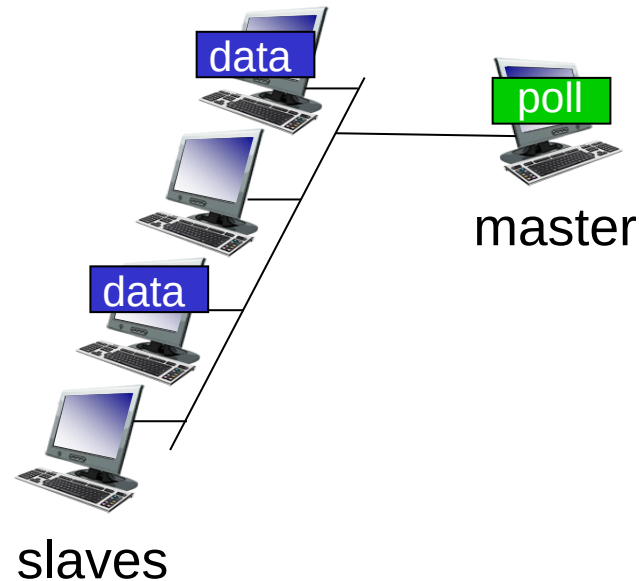
“taking turns” protocols

look for best of both worlds!

“Taking turns” MAC protocols

polling:

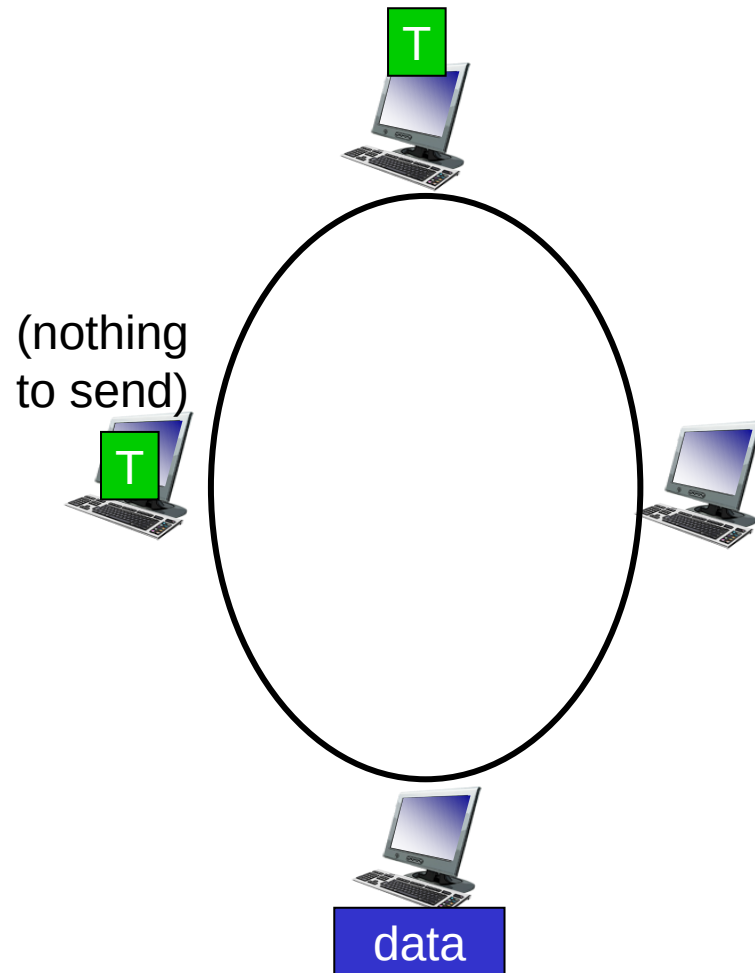
- ❖ master node “invites” slave nodes to transmit in turn
- ❖ typically used with “dumb” slave devices
- ❖ concerns:
 - polling overhead
 - latency
 - single point of failure (master)



“Taking turns” MAC protocols

token passing:

- ❖ control *token* passed from one node to next sequentially.
- ❖ token message
- ❖ concerns:
 - token overhead
 - latency
 - single point of failure (token)



Summary of MAC protocols

- ❖ *channel partitioning*, by time, frequency or code
 - Time Division, Frequency Division
- ❖ *random access* (dynamic),
 - ALOHA, S-ALOHA, CSMA, CSMA/CD
 - carrier sensing: easy in some technologies (wire), hard in others (wireless)
 - CSMA/CD used in Ethernet
 - CSMA/CA used in 802.11
- ❖ *taking turns*
 - polling from central site, token passing
 - bluetooth, FDDI, token ring

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- addressing, ARP
- Ethernet

MAC addresses and ARP

❖ 32-bit IP address:

- *network-layer* address for interface
- used for layer 3 (network layer) forwarding

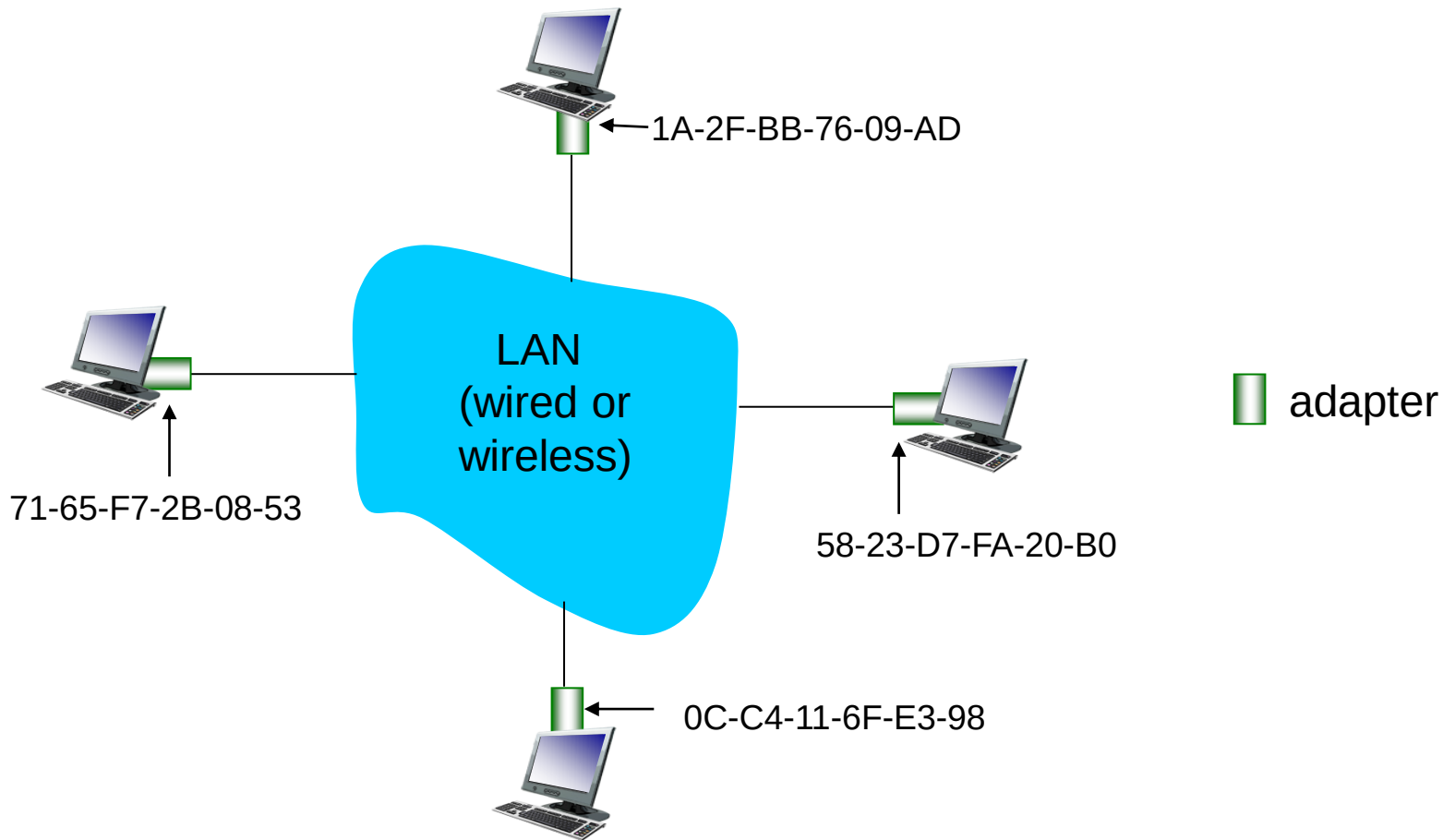
❖ MAC (or LAN or physical or Ethernet) address:

- function: *used ‘locally’ to get frame from one interface to another physically-connected interface (same network, in IP-addressing sense)*
- 48 bit MAC address (for most LANs) burned in NIC ROM, also sometimes software settable
- e.g.: 1A-2F-BB-76-09-AD

hexadecimal (base 16) notation
(each “number” represents 4 bits)

LAN addresses and ARP

each adapter on LAN has unique **LAN** address

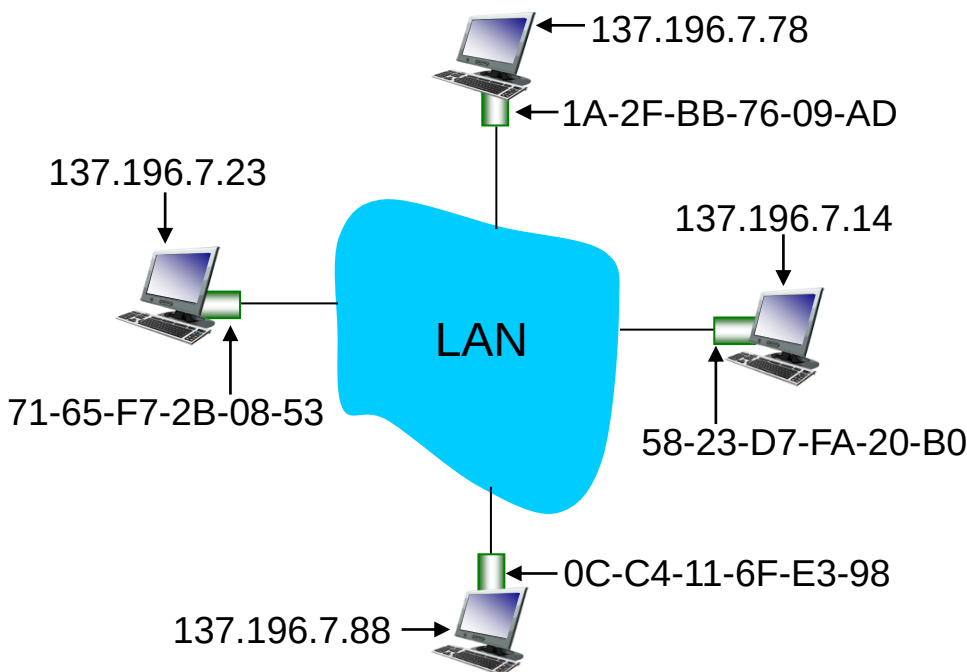


LAN addresses (more)

- ❖ MAC address allocation administered by IEEE
- ❖ manufacturer buys portion of MAC address space (to assure uniqueness)
- ❖ analogy:
 - MAC address: like Social Security Number
 - IP address: like postal address
- ❖ MAC flat address → portability
 - can move LAN card from one LAN to another
- ❖ IP hierarchical address *not* portable
 - address depends on IP subnet to which node is attached

ARP: address resolution protocol

Question: how to determine interface's MAC address, knowing its IP address?



ARP table: each IP node (host, router) on LAN has table

- IP/MAC address mappings for some LAN nodes:
< IP address; MAC address; TTL >
- TTL (Time To Live): time after which address mapping will be forgotten (typically 20 min)

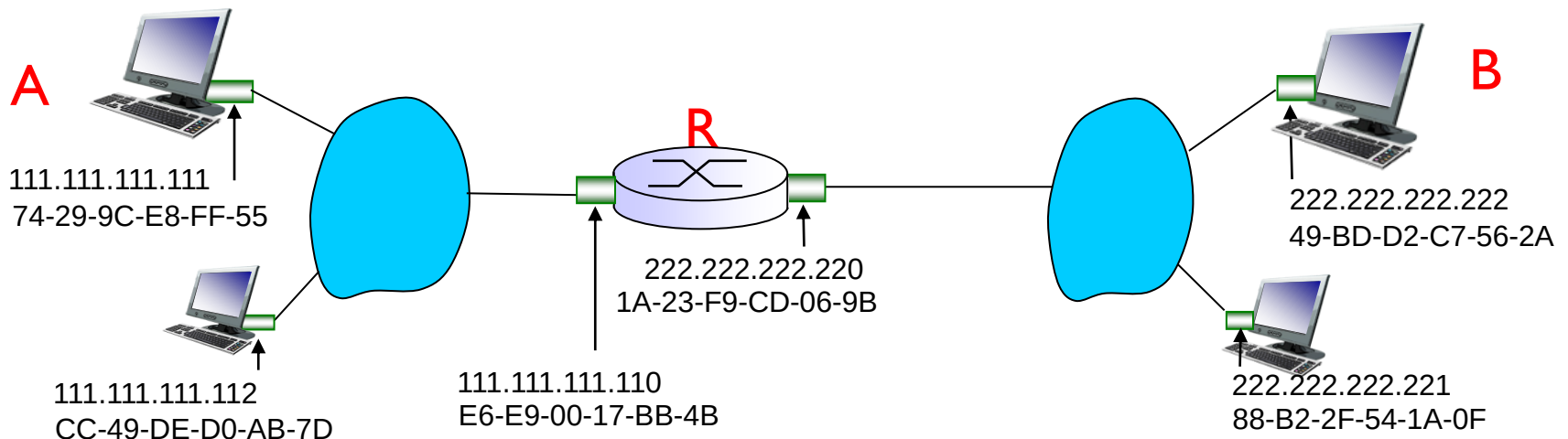
ARP protocol: same LAN

- ❖ A wants to send datagram to B
 - B's MAC address not in A's ARP table.
- ❖ A **broadcasts** ARP query packet, containing B's IP address
 - dest MAC address = FF-FF-FF-FF-FF-FF
 - all nodes on LAN receive ARP query
- ❖ B receives ARP packet, replies to A with its (B's) MAC address
 - frame sent to A's MAC address (unicast)
- ❖ A caches (saves) IP-to-MAC address pair in its ARP table until information becomes old (times out)
 - soft state: information that times out (goes away) unless refreshed
- ❖ ARP is “plug-and-play”:
 - nodes create their ARP tables *without intervention from net administrator*

Addressing: routing to another LAN

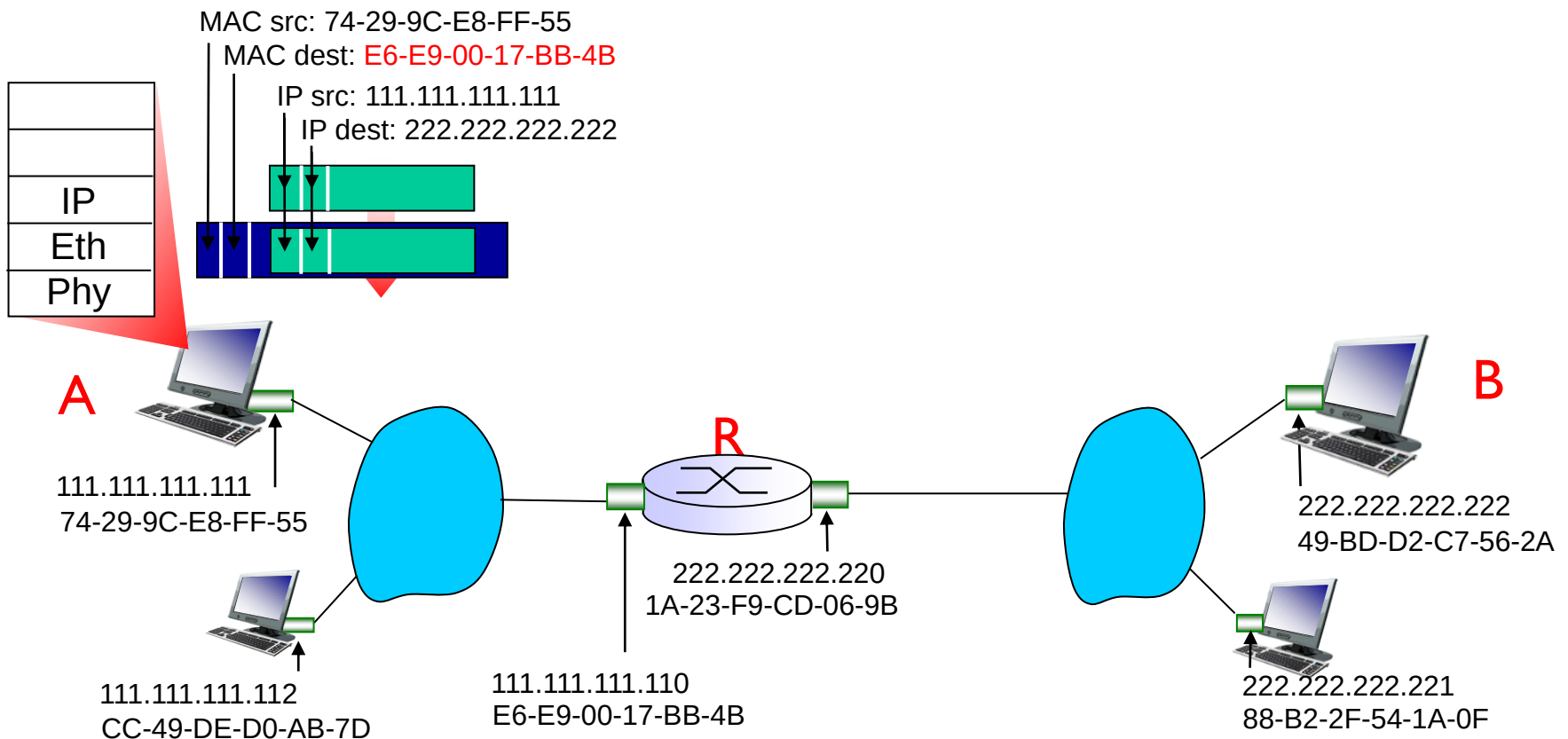
walkthrough: **send datagram from A to B via R**

- focus on addressing – at IP (datagram) and MAC layer (frame)
- assume A knows B's IP address
- assume A knows IP address of first hop router, R (how?)
- assume A knows R's MAC address (how?)



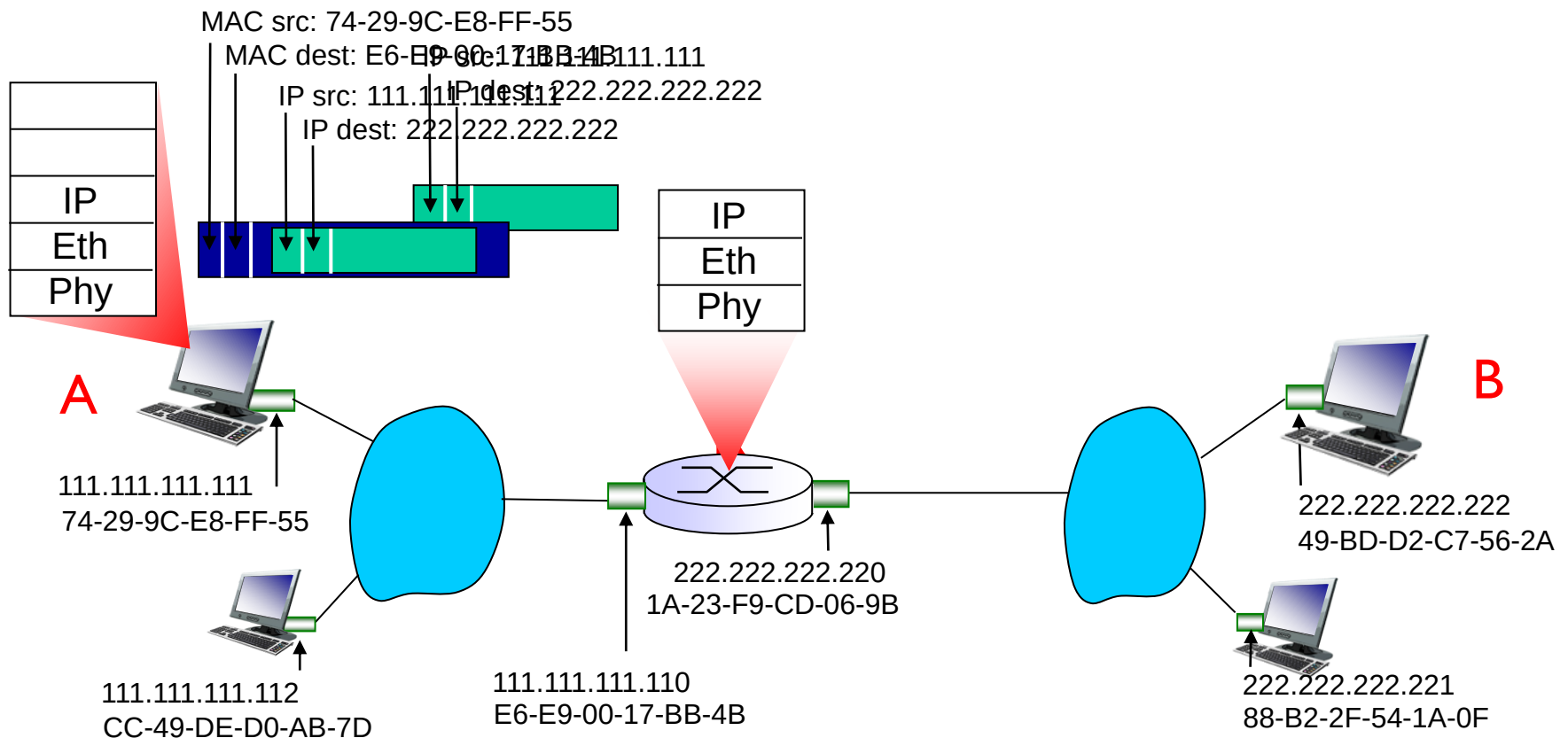
Addressing: routing to another LAN

- ❖ A creates IP datagram with IP source A, destination B
- ❖ A creates link-layer frame with R's MAC address as dest, frame contains A-to-B IP datagram



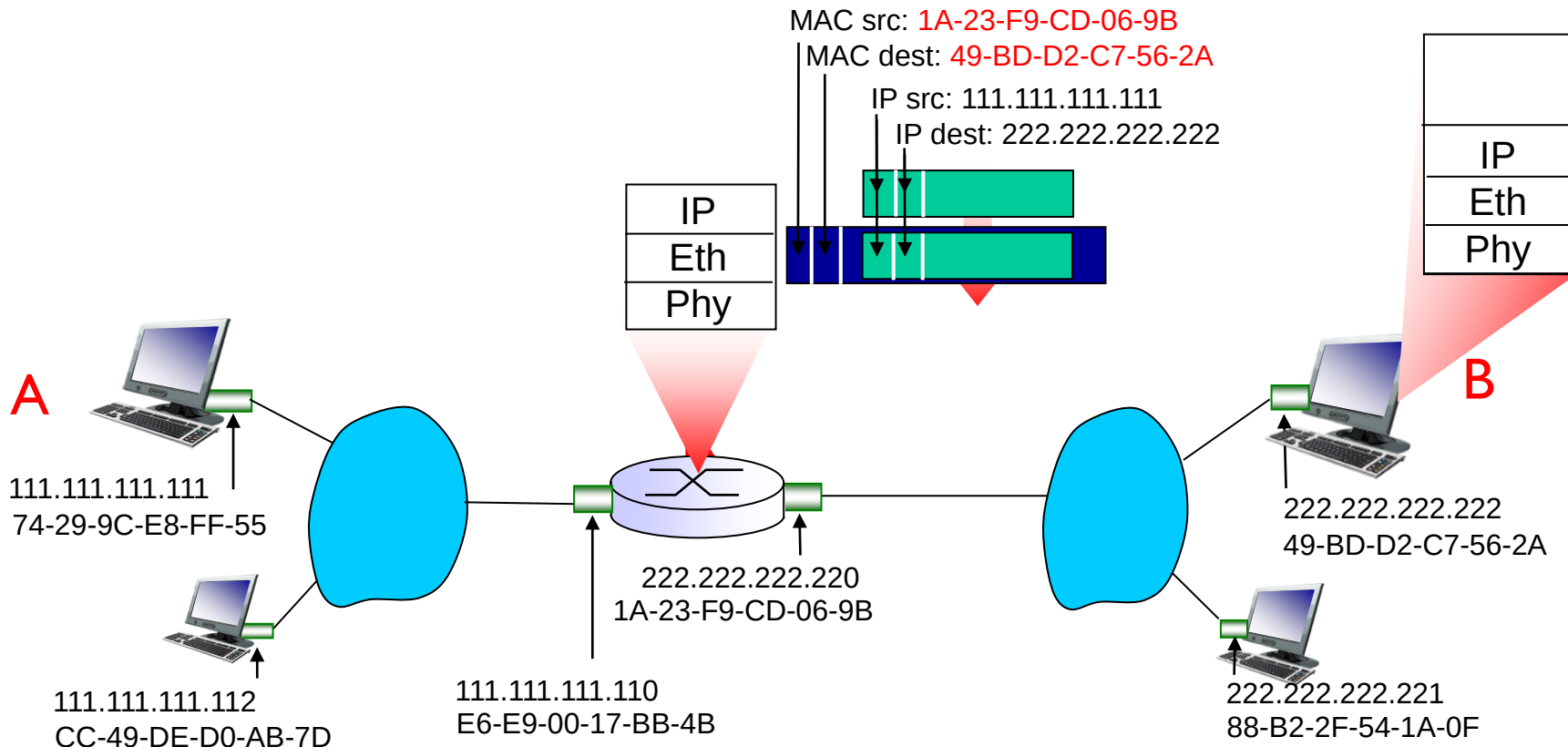
Addressing: routing to another LAN

- ❖ frame sent from A to R
- ❖ frame received at R, datagram removed, passed up to IP



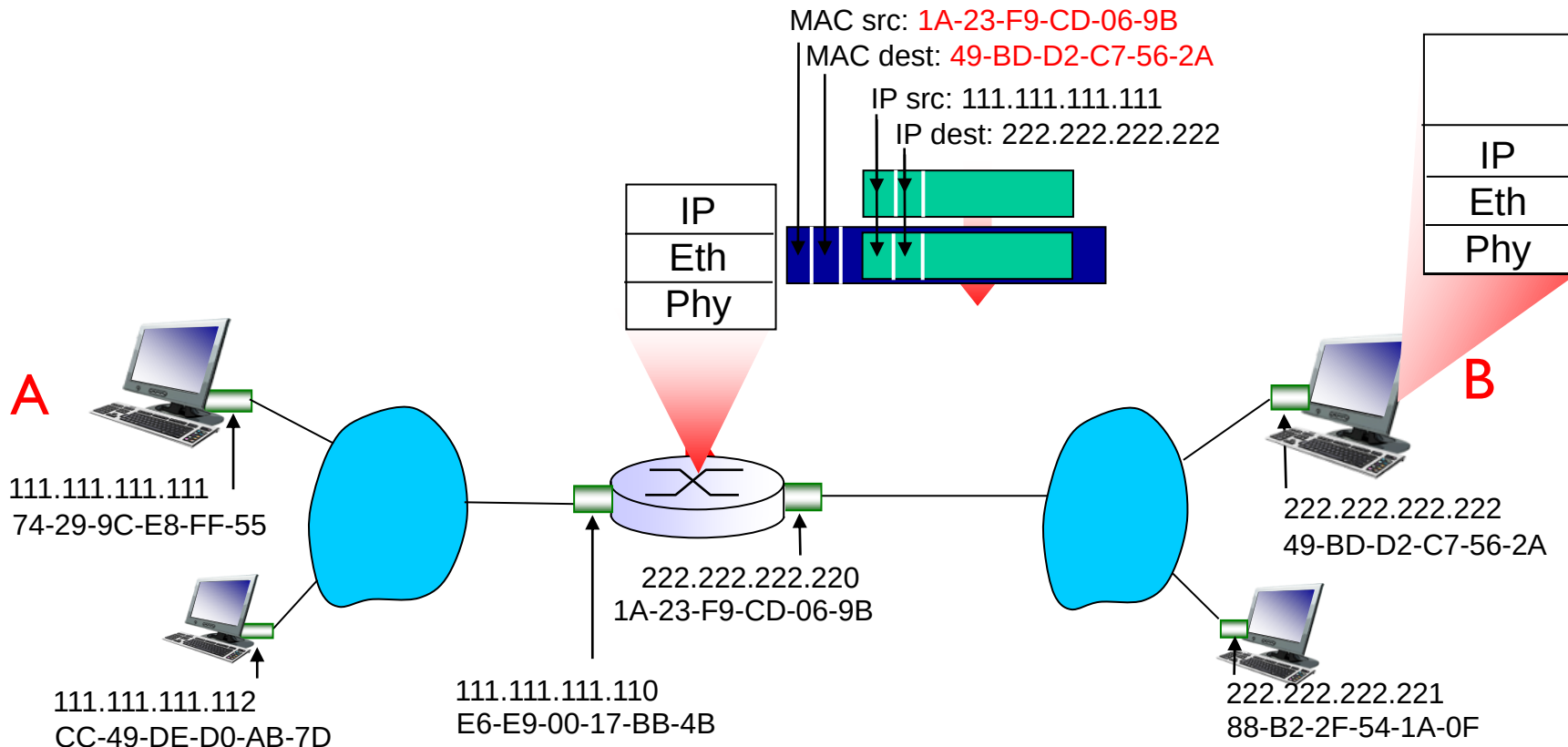
Addressing: routing to another LAN

- ❖ R forwards datagram with IP source A, destination B
- ❖ R creates link-layer frame with B's MAC address as dest, frame contains A-to-B IP datagram



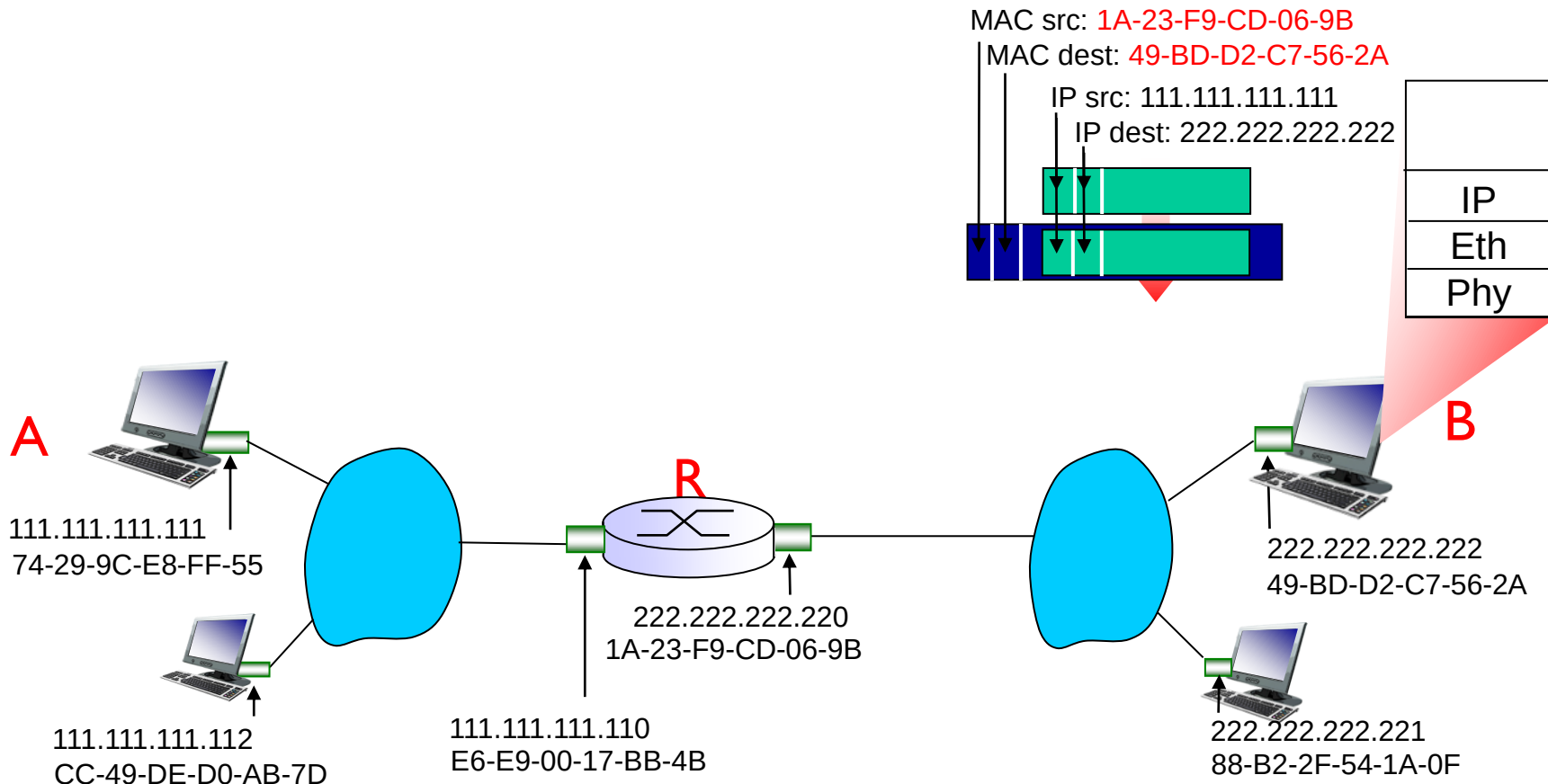
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Addressing: routing to another LAN

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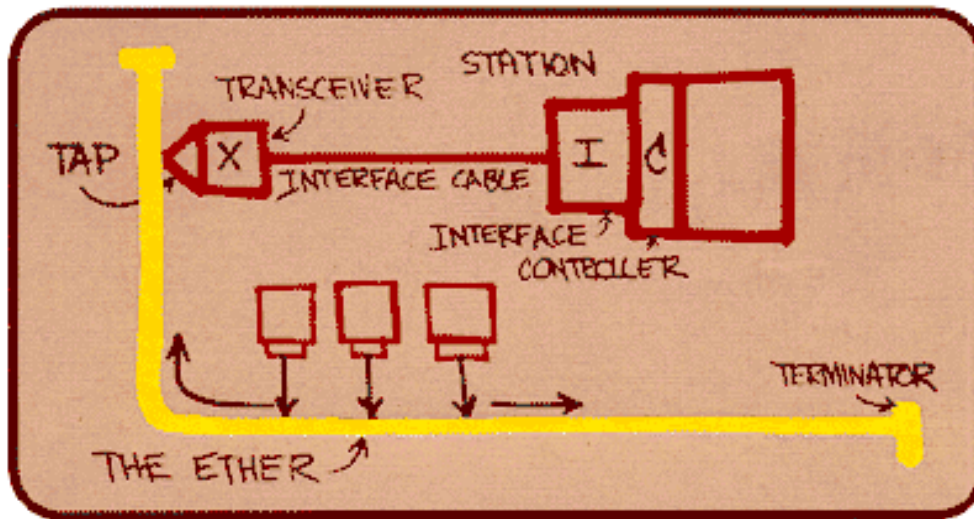
5.4 LANs

- addressing, ARP
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Ethernet

“dominant” wired LAN technology:

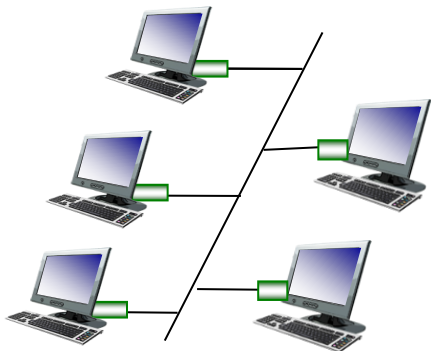
- ❖ cheap \$20 for NIC
- ❖ first widely used LAN technology
- ❖ simpler, cheaper than token LANs and ATM
- ❖ kept up with speed race: 10 Mbps – 10 Gbps



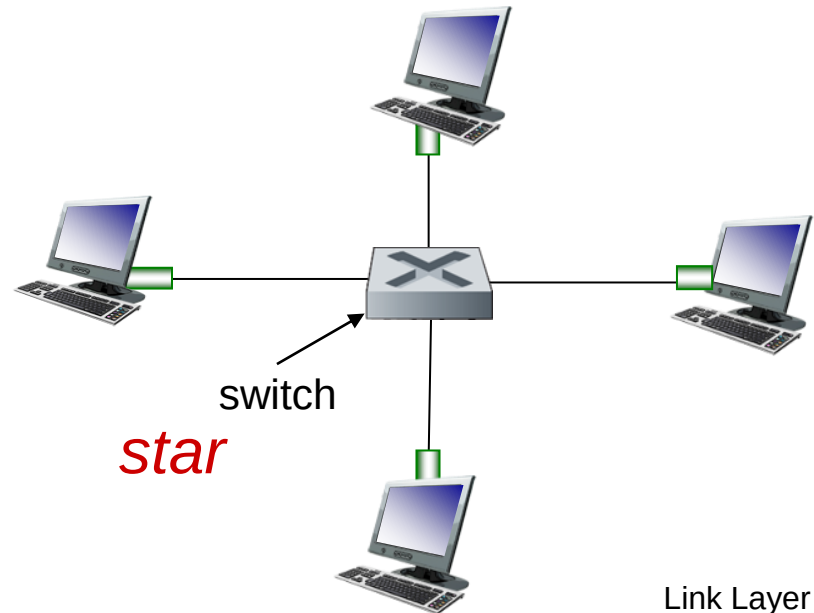
Metcalfe's Ethernet sketch

Ethernet: physical topology

- ❖ *bus*: popular through mid 90s
 - all nodes in same collision domain (can collide with each other)
- ❖ *star*: prevails today
 - active *switch* in center
 - each “spoke” runs a (separate) Ethernet protocol (nodes do not collide with each other)



bus: coaxial cable



Ethernet frame structure

sending adapter encapsulates IP datagram (or other network layer protocol packet) in **Ethernet frame**



preamble:

- ❖ 7 bytes with pattern 10101010 followed by one byte with pattern 10101011
- ❖ used to synchronize receiver, sender clock rates

Ethernet frame structure (more)

- ❖ **addresses:** 6 byte source, destination MAC addresses
 - if adapter receives frame with matching destination address, or with broadcast address (e.g. ARP packet), it passes data in frame to network layer protocol
 - otherwise, adapter discards frame
- ❖ **type:** indicates higher layer protocol (mostly IP but others possible, e.g., Novell IPX, AppleTalk)
- ❖ **CRC:** cyclic redundancy check at receiver
 - error detected: frame is dropped



Ethernet: unreliable, connectionless

- ❖ *connectionless*: no handshaking between sending and receiving NICs
- ❖ *unreliable*: receiving NIC doesn't send acks or nacks to sending NIC
 - data in dropped frames recovered only if initial sender uses higher layer rdt (e.g., TCP), otherwise dropped data lost
- ❖ Ethernet's MAC protocol: unslotted *CSMA/CD with binary backoff*

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